METR — MODEL EVALUATION & THREAT RESEARCH

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every ~7 Months

50% TASK-COMPLETION TIME HORIZON

GPT-2: ~2 seconds (2019) → o3: ~1 hour 50 min (2025) | R^2 = 0.97

Thomas Kwa, Ben West, Joel Becker et al. | arXiv:2503.14499 | March 2025

A New Metric: 50% Task-Completion Time Horizon

A rigorous 3-step methodology yields an intuitive, human-grounded autonomy metric
170 tasks · 800+ human baselines · 2,529 hours · logistic model fit

1 Assemble Task Suite

170 diverse software tasks

Tasks spanning from 1 second to 8 hours of skilled human effort

HCAST · RE-Bench · SWAA suites
3 sources · diverse domains
Shell scripts → CUDA kernels

2 Measure Performance

800+ human baselines collected

Time skilled professionals on each task; run AI agents on same tasks

2,529 total human hours logged
Record per-task pass/fail for AI
Multiple models evaluated

3 Fit Logistic Model

50% success point = time horizon

Fit logistic curve to (task_duration, success_rate) data per model

Read off duration at P(success)=0.5
Compare across model releases
Track doubling time over 6 years

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Figure 2: Methodology overview — task suite assembly, human baseline collection, and logistic model fitting

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

3 curated task suites — HCAST, RE-Bench, SWAA — validated by 800+ human baselines
Coverage: 3 sec to 8 hrs of skilled human effort · diverse domains · 2,529 hrs total baseline time

Task Suite Composition

Suite	Tasks	Duration Range	Description
HCAST	**97**	1 min – 30 hrs	Diverse software tasks
RE-Bench	**7**	~8 hrs each	Difficult ML research engineering
SWAA	**66**	1 sec – 30 sec	Short single-step atomic actions
TOTAL	**170**	3 sec – 8 hrs	800+ human baselines · 2,529 hrs

Example Tasks — From Simplest to Most Complex

Task Name	Human Time	Description
find_shell_script	**3 seconds**	Multiple choice: "Which file is a shell script?"
wikipedia_research	**1 minute**	Research simple factual information from Wikipedia
oxdna_simple	**9 minutes**	Detect & fix bug in molecular dynamics input files
munge_data	**56 minutes**	Write Python script to transform JSON by inferring rules
cuda_backtesting	**8 hours**	Implement custom CUDA kernels for 30× speedup

Complexity →

3 sec ~1 hour 8 hrs

170
Total tasks across 3 suites
HCAST (97) + RE-Bench (7) + SWAA (66)

800+
Human baseline completions
2,529 total hours of human effort logged

3 seconds to 8 hours
9,600× range of human task duration

Benchmark Design Rationale
Human times set task "difficulty" scale.
AI success rate measured independently.
Logistic fit finds 50% success threshold.

AI Task Horizon: 2 Seconds (GPT-2, 2019) → ~1 Hour 50 Minutes (o3, 2025)

AI task-completion horizon has doubled every ~207 days since 2019 — R^2 = 0.97 across 6 years

Exponential fit · 10 frontier models · 2019–2025 · possible acceleration since 2023

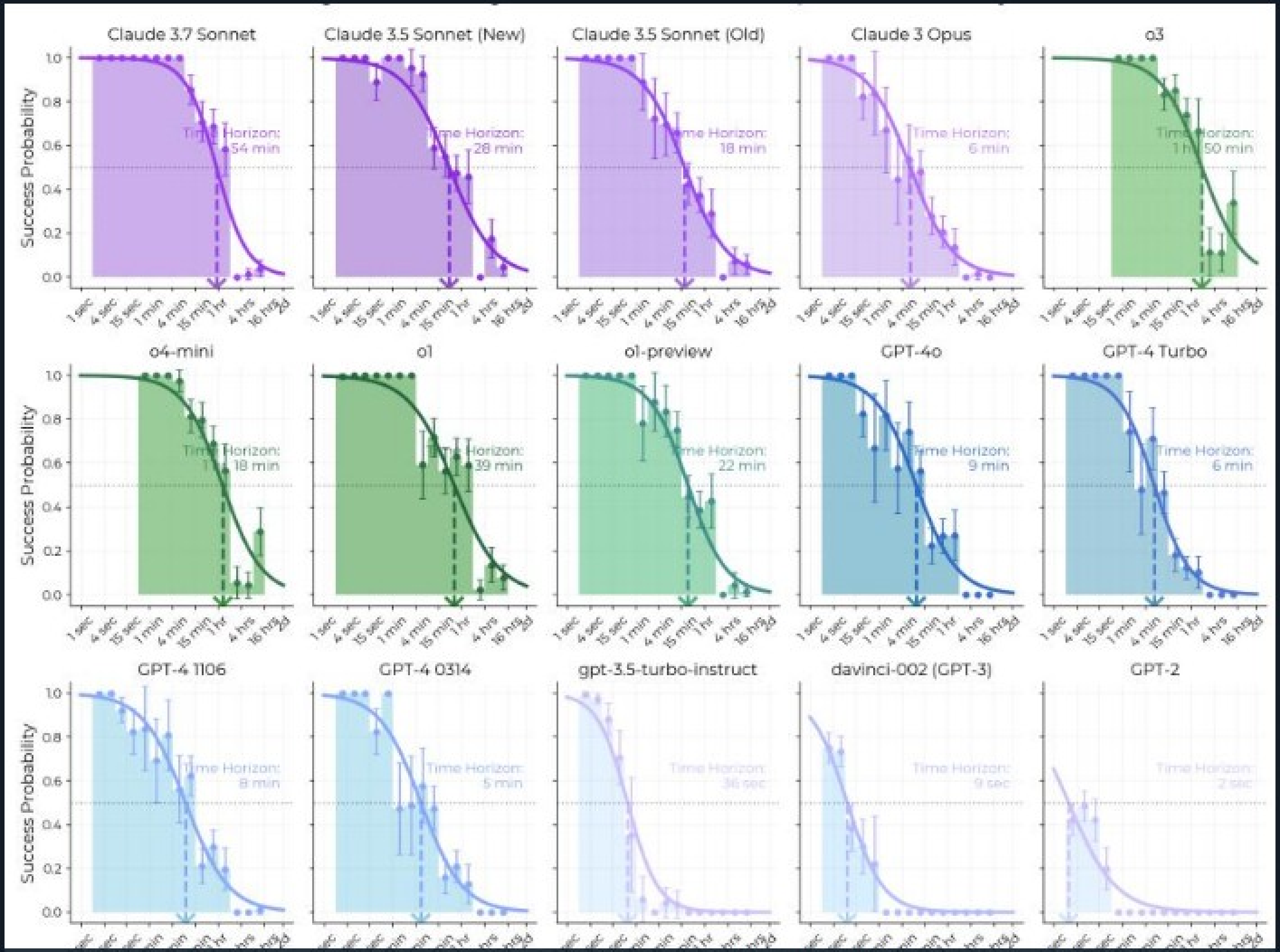

Figure 1: 50% task-completion time horizon vs. model release date (2019–2025) with exponential best-fit line

DOUBLING TIME

~207 days

≈ every 7 months | R^2 = 0.97

Model Milestones

2019 — GPT-2 **~2 seconds**
Baseline · Starting point

2020 — GPT-3 (davinci-002) **~9 seconds**
4.5× improvement

2023 — GPT-3.5-turbo **~36 seconds**
4× further gain

2023 — GPT-4 (0314) **~5 minutes**
8× jump — reasoning milestone

2024 — o1 **~39 minutes**
~5× above GPT-4

2025 — Claude 3.7 Sonnet **~54 minutes**
Rapid iteration cycle

2025 — o3 (Current frontier) **~1 hr 50 min**
3,300× above GPT-2 | 6 years of growth

Possible Acceleration Signal

2023–2025 rate ~20% faster than the 2019–2025 overall baseline rate

Success Probability Drops Sharply as Task Length Increases for All Models

Every model's curve drops with task length — but frontier models shift the curve rightward by orders of magnitude

Logistic success curves · S-curve rightward shift · 80% horizon ≈ 5× shorter than 50% horizon for all models

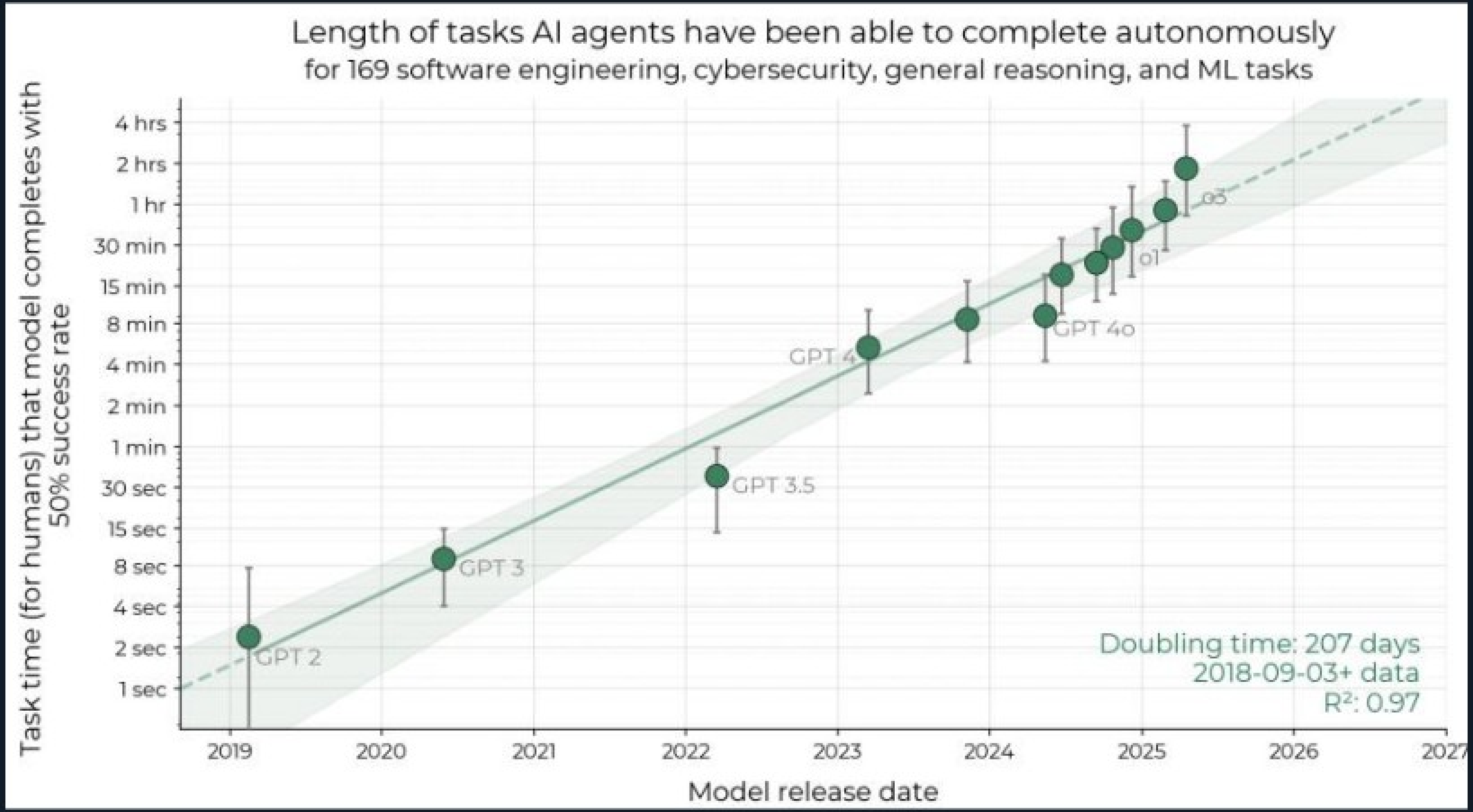

Figure 3: Per-model success probability vs. task duration — each curve shows P(success) as a function of human task length

50% Horizon by Model

Where P(success) = 0.50

Model	Year	50% Horizon
GPT-2	2019	~2 sec
GPT-3 (davinci-002)	2020	~9 sec
GPT-3.5-turbo-instruct	2023	~36 sec
GPT-4 (0314)	2023	**~5 min**
GPT-4o	2024	~9 min
Claude 3 Opus	2024	~6 min
Claude 3.5 Sonnet (New)	2024	~28 min
o1	2024	~39 min
Claude 3.7 Sonnet	2025	**~54 min**
o3 (Current Frontier)	2025	**~1 hr 50 min**

80% Horizon Comparison

≈ 5× shorter than 50% horizon

Same exponential trend, consistent ratio

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

Four capability drivers underpin the exponential trend — but unstructured tasks remain a hard limitation

Logical reasoning · tool use · reliability / self-awareness · error recovery | External validity uncertain

Improved Logical Reasoning

Models chain multi-step deductions with far greater reliability across longer task sequences, maintaining coherent plans.

Enables structured decomposition of complex goals without losing track over multi-hour workflows

→ Foundational driver — all other gains build on this

Better Tool Use

Effective use of shell, code execution, file systems, and web search reduces need for manual intervention steps.

Models can run, test, debug, and iterate autonomously within a single task session

→ Directly extends the time horizon ceiling

Reliability & Self-Awareness

Models better detect when off-track and self-correct before errors compound into unrecoverable task failures.

Calibrated uncertainty prevents confident mistakes from silently derailing long tasks

→ Reduces catastrophic failure rate on long tasks

Error Recovery

Ability to adapt to mistakes mid-task enables completion of longer, complex workflows despite unexpected obstacles.

Iterative refinement of approach rather than irreversible failure when first attempt fails

→ Enables the "long tail" of task completion

Key Limitations

Performance is notably lower on less structured, "messier" tasks than on well-specified benchmarks.

External validity remains an open question — tasks may not perfectly represent the average segment of real-world intellectual labor.

Supplementary experiments found little evidence that trends are slower on more realistic tasks, but uncertainty persists.

Extrapolation: Month-Long Task Automation by 2028–2031

If the trend continues, AI reaches a 1-month task horizon (167 work hours) between mid-2028 and mid-2031
Doubling every ~7 months · 2023–2025 rate ~20% faster · implications for autonomous software development

CURRENT FRONTIER (o3, 2025)
~2 hours
of skilled human work | 3,300× GPT-2

DOUBLING EVERY
~7 months
207 days · $R^2 = 0.97$ · 6 years of data

AUTOMATION THRESHOLD
1 month
≈ 167 work hours · many software projects

Projected Trajectory

Forward Extrapolation — What the Trend Implies

At the current doubling rate of ~207 days, AI will reach a 1-month time horizon (~167 work hours) somewhere between mid-2028 and mid-2031.

This would imply autonomous AI agents capable of completing entire software projects that currently require weeks to months of skilled human engineering work.

The 2023–2025 growth rate is ~20% faster than the 2019–2025 baseline — possible acceleration that could pull the 2028 scenario earlier.

Caveats & Uncertainty

- Extrapolation assumes exponential growth continues — not guaranteed
- Benchmark tasks may not perfectly represent real-world labor distribution
- Safety measures, architectural limits, or regulatory changes may alter path
- Frontier safety governance is urgently

Strategic Implication for AI Safety & Policy

Month-long autonomous task capability would represent a threshold requiring proactive governance frameworks — not reactive responses after the fact.

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

Five key findings demand attention from AI safety researchers, lab leaders, and policymakers — now

R^2=0.97 trend · doubling every 7 months · 2 hrs frontier today · 1 month projected by 2028–2031

1 Robust, Intuitive Metric Established

The 50% task-completion time horizon is a rigorous, human-grounded autonomy metric with R^2=0.97 across 6 years of frontier model releases — enabling systematic tracking of autonomous AI capability over time.

2 Doubling Every ~7 Months — Possibly Accelerating

The ~207-day doubling time holds remarkably consistently from 2019–2025. The 2023–2025 sub-period shows a rate ~20% faster than the overall trend, suggesting the pace of improvement may be increasing.

3 Current Frontier: ~2 Hours of Human-Equivalent Work (o3, 2025)

o3 can now autonomously complete tasks that take skilled professionals nearly 2 hours — up from 2 seconds for GPT-2 in 2019. A 3,300× increase in autonomous task capability in just 6 years.

4 Month-Long Task Automation Projected by 2028–2031

Naive extrapolation predicts AI will reach a 1-month time horizon (~167 work hours) between mid-2028 and mid-2031, implying automation of many month-long software projects if the trend continues.

5 Key Limitations: Unstructured Tasks and External Validity

Performance drops on less structured, "messier" tasks. External validity to real-world job distributions remains uncertain. Governance frameworks must account for both capability level and the trend trajectory.

Safety Governance Imperative

The exponential trend and its acceleration signal that proactive safety governance frameworks — evaluation standards, deployment thresholds, autonomy limits — must be established now, not in 2028.